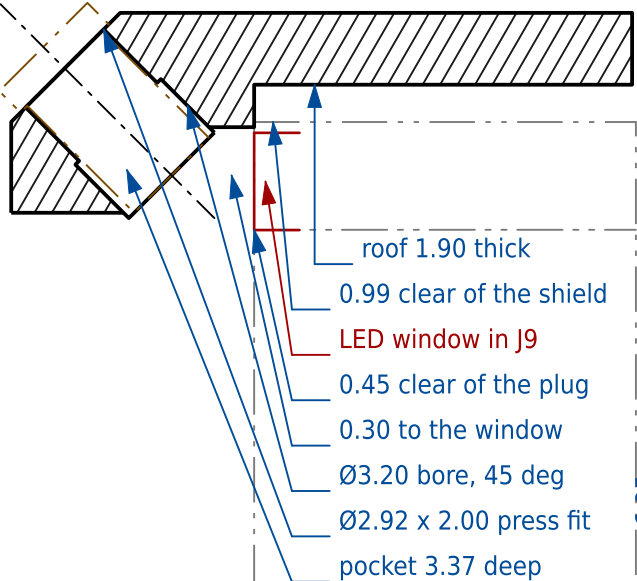
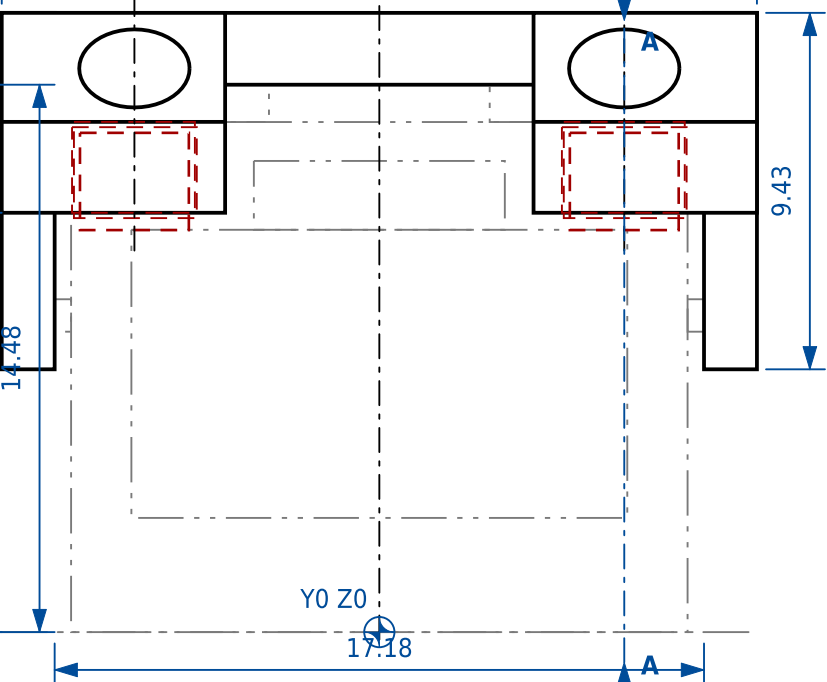


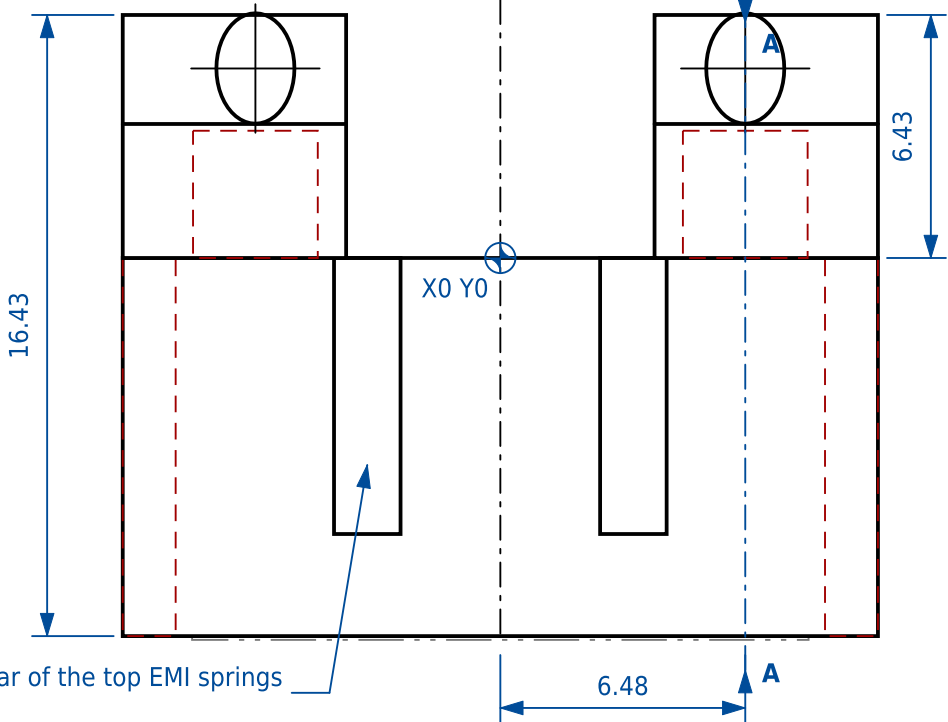
SECTION A-A



FRONT ELEVATION



PLAN



2 slots 1.76 x 7.30, clear of the top EMI springs

NOTES

1. First angle; SECTION A-A is on the right-hand bore's axis. Origin on the jack's centre plane, in its front face, at the board's top surface: X back, Y across, Z up. Symmetric about Y = 0.
2. Two Bivar PLP2-4MM pipes in clear polycarbonate, pressed in from the facet until the flange seats. Each lens looks up and forward at 45 degrees, so a camera looking straight down sees it obliquely.

NOTES (continued)

3. J9's LED windows stand 0.64 mm proud of the face the cheeks land on, so the pipe tips are set from the window, at maximum material.
4. Nothing holds the adapter on but J9's own side EMI springs, 1.4 +/-0.51 proud of a 16.31 +/-0.254 shield, bearing on the skirts: deflected 0.33 mm at minimum material and 1.6 at maximum, the shield never touching.
5. Print upside down, on the top face: the facets and the bores are then 45 degree overhangs and nothing needs support.

BORE SCHEDULE, DERIVED

ID	Y mm	TIP X / Z mm	DIA mm	PRESS mm	PRESS LEN	DEG
B1, B2	+/-6.48	-2.18 / 12.08	3.20	2.92	2.00	45

THE JACK, J9 08B0-1X1T-36-F

FEATURE	Y mm	Z mm	READ AS
LED window, 0.64 proud	5.04..7.92	10.63..13.20	MEASURED
Plug aperture	+/-6.56	3.02..10.64	MEASURED
Latch keyway, widest	+/-3.32	10.64..12.46	MEASURED
Shield	+/-8.15	0..13.49	STATED
EMI spring, side	+/-9.18	7.95..8.80	MEASURED
EMI spring, top	+/-2.92..4.12	to 14.52	MEASURED

LEGEND

- The printed part
- - - Hidden detail: bores and the pocket
- The RJ45 J9 and its EMI springs, adjacent parts
- - - The two Bivar light pipes, bought

NOTES (continued)

6. The press fit is Bivar's Ø2.92 +0.08/-0.05 and that governs, not the general tolerance; ream it if the printed hole comes out undersize.
7. A plug and a plain boot pass under the cheeks; a snagless boot standing proud of the plug's own top face within 7 mm of the jack will foul them.
8. STATED is Bel's own dimension, +/-0.254; MEASURED is read off its geometry, +/-0.20; DERIVED is worked out from those and Bivar's pipe.

SOURCES

1. Board schematic: https://digilent.com/reference/_media/reference/programmable-logic/arty-a7/arty_a7_sch.pdf - Digilent, Arty A7 rev E.0, sheet 8 of 12 (ETHERNET): J9 is 08B0-1X1T-36-F.
2. Jack drawing: <https://www.belfuse.com/media/drawings/products/magjack%20ICMs/dr-mag-08b0-1x1t-36-f.pdf> - Bel 08B01X1T36-F rev E; MEASURED figures are scaled off its front and side views.
3. Board drawing: https://digilent.com/reference/_media/reference/programmable-logic/arty-a7/arty_a7.zip - Digilent, Arty_A7_DXF.DXF: board locks 16.104 apart against Bel's 16.13; centre plane y = 44.0, and through Bel's 7.75 the front face x = -0.101 +/-0.628, which is how far Digilent's own plot of the jack disagrees.
4. Light pipe drawing: https://www.bivar.com/parts_content/Datasheets/PLP2-XXX.pdf - Bivar PLP2-XXX rev Y: Ø2.8 pipe on Ø3.1 ribs, Ø3.3 flange, Ø0.115 in hole, 0.047-0.093 in panel.

ARTY A7 ETHERNET LIGHT PIPE

Clips over J9 and carries its two LEDs to a face you can see from above

DRAWN generated		VERSION v0.0-468-g7b8e30e	UNITS mm	SCALE 5:1
SIZE A3	SHEET 1 OF 1	DRAWING NO FPGA-LP-ARTY		REV A
MATERIAL 3D printed, opaque		GENERAL TOLERANCE printed +/-0.20 bores: see note on the press fit		PROJECTION